

Behnush Kiani

Regina, SK, Canada | (306) 880- 4230 |Email: Behnushkiani@gmail.com| [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

MSc graduate in Computer Science with a background in machine learning, data science, statistical modeling, and deep learning. Experienced in building ML pipelines, predictive models, feature engineering, and statistical analysis using Python and SQL. Hands-on experience with enterprise data analytics, computer vision, deep learning, and LLM/RAG applications through academic research and practical AI projects.

TECHNICAL SKILLS

ML Systems & Modeling

- ML pipeline development: data preprocessing, feature engineering, model training, evaluation, and deployment
- Supervised & unsupervised learning: classification, regression, clustering, ranking, anomaly detection
- Computer vision: CNNs (ResNet, VGG), image similarity, pattern recognition, OpenCV
- Understanding of LLM concepts, prompt engineering, Retrieval-Augmented Generation (RAG), embeddings, and vector search.

Engineering & Infrastructure

- Python (primary), SQL, PyTorch, TensorFlow, scikit-learn, pandas, NumPy, OpenCV
- Structured and unstructured data processing: metadata, images, text descriptions
- ERP systems, enterprise data management, large-scale dataset handling

WORK EXPERIENCE

Graduate Research & Teaching Assistant

University of Regina — Regina, Canada | 2023 –2026

- generative parameter estimation for fractal images using pretrained convolutional neural networks.
- Investigated the effectiveness of transfer learning and CNN-based regression for estimating continuous fractal-generation parameters from image data.
- Evaluated and compared model performance through cross-validation, statistical analysis, R^2 , and MSE, analyzing the impact of model architecture and image characteristics on prediction performance
- Assisted students with Machine Learning, Data Science, Python, and AI-related coursework and projects.

Data Analyst

Hamkaran System (System Group) — Tehran, Iran | 2017 – 2022

- Built machine learning and analytics solutions across 5 ERP business domains, supporting 50+ enterprise clients in sales, finance, inventory, customer, and operations.
- Built a customer behavior prediction model using historical sales and ERP data to identify purchasing patterns and improve demand forecasting, resulting in an 18% improvement in sales prediction accuracy.
- Developed and evaluated classification, regression, clustering, and anomaly-detection models to identify business risks, customer patterns, operational issues, and unusual behavior in enterprise ERP data.
- Analyzed large-scale ERP datasets using Python and SQL, performing correlation analysis, exploratory data analysis (EDA), and statistical analysis to identify relationships between variables, trends, anomalies, and data-quality issues across 50+ enterprise customer environments.

- Automated recurring data preparation, validation, and reporting workflows, reducing manual processing effort by approximately 25% while improving the consistency, efficiency, and reliability of data analysis.
- Collaborated with ERP consultants, technical teams, and business stakeholders across 5 core business areas to translate business requirements into data and machine learning solutions and communicate actionable insights.

PROJECTS

Fractal Image Parameter Estimation | PyTorch, ResNet, VGG

Built and published an original synthetic fractal image dataset of 2,000 labeled images with generation parameters, creating the full data pipeline from image generation and labeling through preprocessing and model training. Fine-tuned pretrained ResNet-18 and VGG-16 models to estimate fractal-generation parameters directly from images, evaluating performance using R^2 and MSE.

Agentic AI Platform for Workplace Safety & Document Analysis | YOLOv8, OCR, LLMs

Built an AI-powered multi-tool system that analyzes workplace safety images and business documents using object detection, OCR, text classification, and LLM-based routing. Trained a YOLOv8 PPE detection model to identify safety equipment such as helmets and vests and combined the results with document analysis and automated risk reporting.

Late Delivery Risk Prediction | XGBoost, LightGBM, Optuna, SHAP

Built an end-to-end machine learning pipeline to predict the risk of late deliveries from operational and supply-chain data. Trained and optimized XGBoost and LightGBM models using Optuna and used SHAP to explain the key factors driving delivery-risk predictions.

EDUCATION

Master of Science (M.Sc.) in Computer Science

University of Regina, Regina, Saskatchewan, Canada | Graduated: 2026

Master of Science (M.Sc.) in Electronic Commerce

Islamic Azad University, Science and Research Branch, Tehran, Iran | Graduated: 2016

Bachelor of Science (B.Sc.) in Software Engineering

Islamic Azad University, Qazvin, Iran | Graduated: 2014